



FREE GUIDE · FOR COACHES WHO WANT TO COACH NUTRITION PROPERLY

Nutrition Coaching Fundamentals for PTs

The evidence behind energy balance, protein and macros, which supplements are actually worth recommending, where your scope of practice ends, and how to coach nutrition clients actually stick to.

Who this guide is for

This is written for coaches who already know the basics from their Certificate III/IV: the food groups, a rough idea of macros, and how to run a client through the Australian Dietary Guidelines. This guide goes past that, into the actual research behind the numbers, where the evidence is strong and where popular advice has drifted well past what the science supports, and exactly where your scope of practice as a trainer ends.

Nutrition is also where it's easiest for a well-meaning PT to accidentally do harm: overpromising on a supplement, handing a client an overly rigid meal plan, or quietly practising outside what you're insured and qualified for. This guide treats that as seriously as the science.

A NOTE ON ACCURACY

Nutrition science moves fast and gets oversimplified in the process. Where the evidence is strong and consistent, it's stated plainly. Where it's genuinely mixed, emerging, or weaker than the popular version suggests, that's flagged explicitly rather than smoothed over.

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Energy Balance, Properly Understood

Calories in versus calories out is true as a law of thermodynamics. Where coaching advice goes wrong is treating a client's total daily energy expenditure (TDEE) as a fixed, known number, when it's actually an estimate stacked on top of another estimate: a predictive equation, and self-reported intake that's rarely as accurate as it feels.

Picking an equation, and knowing its error margin

Mifflin–St Jeor is the most accurate predictive equation on average for both non-obese and obese adults, ahead of the older Harris–Benedict equation some courses still teach. Katch–McArdle, which uses lean body mass instead of total weight, is theoretically sharper for very lean or muscular clients, but only as good as the body-fat estimate feeding it, which is rarely precise enough in a gym setting to earn the extra complexity. Even the best equation still carries roughly $\pm 10\text{--}15\%$ individual error, so the real skill isn't picking the “right” formula, it's treating the number as a starting point and adjusting it from two to three weeks of real weight-trend data.

Metabolic adaptation is real, but its size depends on how you diet

When someone diets, resting metabolic rate tends to drop by more than reduced body mass alone predicts, a phenomenon called adaptive thermogenesis. The most cited evidence comes from a six-year follow-up of contestants from an extreme televised weight-loss competition, whose metabolic rate stayed suppressed by roughly 500 calories a day, far beyond what their regained weight explained. That case involved extremely rapid, extreme weight loss and isn't representative of a client losing half a kilo a week. Moderate, sensible dieting produces a smaller effect, generally a few percent beyond what reduced body mass predicts, part metabolic and part simply moving less without realizing it, a drop in incidental daily activity known as NEAT.

Burning fat isn't the same as losing fat

“Burning fat” means fat oxidation, the body using fat as fuel, which happens to some degree all day in everyone. “Losing fat” means a net drop in stored body fat over time, decided by total energy balance across weeks, not which fuel a single session happened to burn. Chase energy balance across the week, not fuel source during a workout.

DIET BREAKS: A REASONABLE TOOL, NOT A PROVEN METABOLIC HACK

Planned higher-calorie weeks give clients real psychological relief and have a plausible hormonal rationale. But direct evidence that they preserve more muscle or metabolic rate than the same total calories eaten continuously is thin and mixed. Use them for adherence, not as a metabolism trick.

Protein: The Numbers That Actually Matter

Of everything in this guide, protein has the clearest, most heavily researched dose-response relationship, which is exactly why it's worth using the real numbers rather than one rounded rule of thumb for every client.

- ✓ **General hypertrophy training:** 1.6–2.2 g/kg of bodyweight per day covers the large majority of trained clients, with research finding little extra muscle gain above roughly 1.6 g/kg for most people.
- ✓ **Cutting or calorie deficit:** 2.3–3.1 g/kg of lean body mass for lean, trained clients actively dieting, meaningfully higher than the maintenance number, because protein needs rise specifically to protect muscle while calories are restricted.
- ✓ **Older adult clients:** 1.0–1.2 g/kg/day at minimum, well above the general 0.8 g/kg RDA, rising to 1.2–1.5 g/kg with illness or injury, and split across meals at 25–30g per meal to overcome age-related anabolic resistance.
- ✓ **General, untrained population:** the standard 0.8 g/kg RDA is a floor for avoiding deficiency, not a target for anyone doing structured training.

The anabolic window is far wider than PT courses suggest

The idea that protein must be consumed within 30–60 minutes of training to “count” doesn't hold up. Once training is consistent, total daily protein intake drives long-term results far more than the precise timing of any single dose. There's a real leucine threshold behind muscle protein synthesis, roughly 2–3g of leucine, found in about 25–40g of most quality protein sources, which is a good reason to spread protein across three to four meals rather than back-loading it into one. That's a meal-distribution principle, not a post-workout emergency.

Does pairing protein with carbs boost muscle protein synthesis?

Adding carbohydrate to a protein dose doesn't measurably increase muscle protein synthesis beyond what adequate protein already provides on its own. Insulin's real role here is suppressing muscle protein breakdown, not adding an extra synthesis boost, a smaller and different benefit than the old “insulin spike builds muscle” idea suggests. Pairing carbs with protein around training is still a perfectly good, practical way to refuel and hit daily targets, just not for the reason many coaches were taught.

PLANT PROTEIN AND THE “COMBINING” MYTH

Plant proteins are typically lower in one or two essential amino acids, but this evens out across the day as different sources are eaten, not within a single meal. Vegetarian and vegan clients don't need to combine proteins at every sitting, they need enough total protein and enough variety across the week.

Carbs, Fat & the Diets That Get Overhyped

Once protein and total calories are matched, most of the diet-style debates that dominate fitness media, low-carb versus low-fat, keto versus balanced, shrink down to personal preference and adherence rather than physiology.

What carbohydrate is actually doing

Carbohydrate's main job is fuelling glycogen stores and supplying glucose to the brain. Very low-carbohydrate approaches can undercut training even while the scale looks fine, a separate question from fat loss: carbohydrate isn't inherently fattening, and cutting it isn't inherently superior once protein and calories are matched.

Match protein and calories, not macronutrient ratio

A 609-person, 12-month randomized trial comparing a healthy low-fat diet to a healthy low-carb diet found no meaningful difference in weight loss between them, and no evidence that certain people respond better to one approach based on genetics or baseline insulin. Smaller trials on ketogenic diets in trained people generally find similar body-composition outcomes to non-ketogenic, calorie-matched diets too. The deciding factor for which style to recommend a client is almost always which one they'll actually follow for months, not which one is metabolically superior.

Fat has a floor, even if the exact number is debated

Very low-fat diets are associated with suppressed reproductive hormones, including testosterone, particularly combined with low overall energy intake. The research hasn't nailed down one universal minimum, but keeping fat above roughly 20% of total calories is a sensible, evidence-informed floor for clients dieting hard, rather than a strict cutoff.

Fibre is one of the best-supported numbers in this guide

Large reviews pooling well over a hundred studies find fibre intakes of 25–30g a day associated with meaningfully lower rates of heart disease, stroke, type 2 diabetes and bowel cancer, with benefits continuing to climb at even higher intakes. Most clients fall well short of this, and it's a higher-leverage conversation than almost any macro debate. Fibre comes in two types: soluble (oats, legumes, fruit), which feeds gut bacteria, and insoluble (whole grains, vegetable skins), which adds bulk, and most clients need more of both.

A USEFUL REFRAME FOR CLIENTS

"Which diet is best" is usually the wrong question. "Which diet can you eat for the next six months without white-knuckling it" predicts real-world results far better.

Meal Timing, Frequency & Fasting

How often and when a client eats gets disproportionate attention relative to how little it actually changes outcomes, once total intake is held constant.

Meal frequency doesn't move metabolism

Eating six small meals instead of three larger ones doesn't measurably raise metabolic rate or burn more calories once daily totals are matched. Meal frequency is a tool for hunger management and fitting a client's schedule, not a metabolic lever.

Intermittent fasting works, but not for the reason it's usually sold

A well-controlled trial on 16:8 time-restricted eating found no significant weight-loss advantage over eating across a normal window, and a small loss of lean mass in the fasting group. Larger recent reviews describe fasting-style approaches as working primarily as a calorie-control tool, an eating window that makes it easier for many people to eat less, rather than a unique fat-burning mechanism. It's a legitimate option for a client who finds it easier to stick to, not a superior default for everyone.

Nutrient timing matters more for elite athletes than for the average client

Precise pre- and post-workout feeding windows are genuinely more important for competitive or high-training-volume athletes than for someone lifting a few times a week. For most PT clients, hitting the day's total protein and calories reliably outperforms any specific timing protocol.

A PRACTICAL TAKEAWAY

If a client's meal pattern fits their life and lets them hit their targets consistently, that pattern is correct for them, regardless of how many meals it involves or when they land.

Hydration & the Supplements Worth Recommending

This is where the fitness industry sells the most and the evidence supports the least. A simple evidence tier makes it easy to know what's actually worth a client's money.

Hydration: forget the fixed number

The old “eight glasses a day” rule isn't grounded in strong evidence. Fluid needs vary hugely by body size, climate, sweat rate and the water content of what someone eats. Thirst-guided drinking covers the large majority of clients; the widely used rule that roughly 2% body-mass loss in fluid starts to measurably hurt exercise performance is a helpful coaching heuristic, not an exact, universally proven threshold.

The evidence tiers, in order

- ✓ **Strong evidence:** creatine monohydrate (3–5g daily, no need to “cycle” it), caffeine (roughly 3–6mg per kg of bodyweight, 30–60 minutes pre-training), and protein powder used simply as a convenient way to hit a protein target.
- ✓ **Promising, not yet proven:** fish oil, curcumin, and most polyphenol supplements. Reasonable to try, not worth overselling.
- ✓ **Weak or no evidence at typical doses:** BCAAs (a complete protein source does the same job, and does it better), most “fat burners”, and isolated amino acids stacked on top of an already-adequate protein intake.
- ✓ **Don't recommend:** prohormones, unregulated “testosterone boosters”, and anything from the banned or high-contamination-risk category, both for legal exposure and because the evidence behind them is essentially absent.

CREATINE IS THE MOST RESEARCHED, SAFEST SUPPLEMENT YOUR CLIENTS AREN'T TAKING

Decades of research support creatine monohydrate for strength, power, and even some cognitive outcomes, at a simple daily dose, with an excellent long-term safety record. It's consistently the highest-value recommendation on this page.

Why Nutrition Plans Fail in the Real World

A textbook-perfect macro plan a client abandons after two weeks has produced worse results than an imperfect plan they actually followed for six months. This chapter is about the gap between correct and sustainable.

Flexible versus rigid dieting

Rigid, all-or-nothing rules, never eating a certain food, treating one meal as a total failure, are consistently linked to a higher risk of disordered eating patterns than a flexible approach built around overall targets and moderation. This matters most for clients already showing perfectionist or anxious tendencies around food, where a strict plan can do real harm rather than simply underperform.

Self-monitoring works, with a caveat

Clients who track their food consistently tend to do better than those who don't, largely because tracking closes the gap between what people think they eat and what they actually eat. The caveat is real: for a minority of clients, especially anyone with a history of disordered eating, food logging can tip into obsessive checking rather than helpful awareness, and is worth watching for.

Habits beat willpower over the long run

Coaching built around specific, repeatable habits, a consistent breakfast, a standing grocery list, a default go-to meal on busy nights, shows real promise for long-term adherence, even though this area has fewer large trials than calorie-counting research does. In practice, one or two habits a client will actually keep tend to outperform a comprehensive plan they'll abandon.

THE REAL SKILL IS HONESTY ABOUT THE TRADE-OFF

Every dietary approach trades off being textbook-optimal against being something a real person, with a real job and a real social life, can sustain. Naming that trade-off out loud with a client is usually more useful than a more precise macro split.

Know Your Lane: Scope of Practice

Nutrition is where it's easiest for a well-meaning PT to unintentionally step outside what they're qualified and insured to do. Knowing the line protects the client, and protects your business.

What you can confidently cover

- ✓ General healthy eating guidance based on the Australian Dietary Guidelines: food groups, portion sizes, and comparing a client's current pattern against them.
- ✓ General weight-management conversations built around energy balance, food quality, and habits, not individualized medical nutrition.
- ✓ Food label literacy, and general examples of balanced meals and snacks.

What belongs with an Accredited Practising Dietitian

- ✓ Personalized meal plans built around a specific medical condition: diabetes, coeliac disease, allergies, kidney or liver disease.
- ✓ Detailed macro or micronutrient prescriptions for clinical or complex cases, and any drug-nutrient interaction question.
- ✓ Specialized or restrictive diet planning. In-depth support for a client transitioning to vegetarian or vegan eating sits closer to the dietitian line than most PTs assume.

RED FLAGS THAT MEAN REFER, DON'T COACH

Any diagnosed medical condition, signs of disordered eating, a client fixated on extreme restriction, or a client simply not progressing despite following general guidance well, are all signals to bring in a dietitian rather than adding more nutrition advice of your own. In Australia, "dietitian" isn't a protected title the way "physiotherapist" is, so look specifically for the Accredited Practising Dietitian (APD) credential through Dietitians Australia when referring.

Coaching Different Clients Differently

The principles above are the floor. The clients most likely to need more than the floor are the ones covered here.

Vegetarian and vegan clients

Protein adequacy is achievable on a well-planned plant-based diet. The real watch-items are vitamin B12 (absent from plant foods entirely), iron and zinc (less bioavailable from plant sources), and creatine, which is essentially absent from a vegan diet. Vegetarian and vegan clients typically start with lower baseline muscle creatine stores, which if anything makes them more responsive to creatine supplementation, not less.

Female clients and energy availability

The modern framework here is Relative Energy Deficiency in Sport (RED-S): not eating enough relative to training volume and body composition, which can affect menstrual function, bone health, metabolic rate and performance in any client, not only competitive female athletes. It's worth actively screening for in anyone training hard while under-eating, especially a client with irregular or absent periods. The idea that appetite and needs shift meaningfully across the menstrual cycle is a genuine, active research area, real for some individuals, but not yet consistent enough to build a rigid nutrition plan around by default.

Older adult clients

Protein needs rise with age, roughly 1.0–1.2 g/kg/day at minimum, more with illness, because ageing muscle becomes less responsive to a given protein dose. Hitting 25–30g at each main meal matters more here than it does for a younger client who can get away with an uneven distribution across the day.

Common Nutrition Coaching Mistakes

Most nutrition advice that quietly underperforms traces back to one or two of these, not a single dramatic error.



Treating a TDEE number as exact

It's a starting estimate with meaningful error built in. Adjust from real weight-trend data, not blind faith in the formula.



Chasing timing over totals

Precise pre- and post-workout windows matter far less than hitting the day's total protein and calories consistently.



Recommending restrictive diets by default

An all-or-nothing plan a client can't sustain does less for them than a flexible one they actually follow.



Selling supplements ahead of the fundamentals

No BCAA, fat burner, or fasting protocol outperforms fixing total protein, total calories and consistency first.



Giving individualized medical nutrition advice

Prescribing meal plans for a medical condition, or advising on medication interactions, is outside a PT's scope. Refer to a dietitian.



Applying one nutrition rule to every client

A vegan client, a perimenopausal client and a 70-year-old client all have genuinely different priorities. The floor is the same; the client in front of you isn't.

The Nutrition Coaching Checklist

Every step from this guide, in order. Print this page or keep it open while you build your next client's plan.

- ✓ Estimate TDEE with Mifflin–St Jeor, then adjust from two to three weeks of real weight-trend data
- ✓ Set protein by goal: 1.6–2.2 g/kg general, 2.3–3.1 g/kg lean mass when cutting, 1.0–1.5 g/kg for older adults
- ✓ Spread protein across three to four meals rather than back-loading it into one
- ✓ Pick a diet style by adherence fit, not by macronutrient ratio, once protein and calories are matched
- ✓ Keep fat above roughly 20% of calories, and push fibre toward 25–30g a day
- ✓ Treat meal frequency and fasting windows as adherence tools, not metabolic levers
- ✓ Recommend fluids by thirst, not a fixed daily volume
- ✓ Lead supplement advice with creatine and caffeine, the strongest evidence in the category
- ✓ Avoid rigid, all-or-nothing plans, especially with perfectionist or anxious clients
- ✓ Watch for food logging tipping into obsessive checking, not just helpful awareness
- ✓ Keep general nutrition advice within scope, and know the red flags that mean referring to a dietitian
- ✓ Watch B12, iron, zinc and creatine specifically for vegetarian and vegan clients
- ✓ Screen for under-eating relative to training load in any client, not only competitive athletes
- ✓ Individualize by the client in front of you, never by a single rule applied to everyone



A note from Totality's founder

"I built Totality because I'm a personal trainer myself, and I was sick of running my business through four apps that didn't talk to each other. This guide is the advice I wish someone had handed me when I started: the practical, unglamorous stuff nobody puts in a course brochure."

Taj Donaldson, Founder & CEO, Totality · Personal Trainer

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